

Getting Familiar *with* Multimedia Streaming Protocols

An understanding of the multimedia application is a must in today's world because there are many products based on it, like the IP phone, IP TV, video conferencing applications, network video recorders, etc. These products have become essentials in this modern era of communication.



Real-Time Protocol (RTP)

RTP is an Internet standard protocol for the transport of real-time data. It is used to exchange video/audio data over a network. RTP does not check bandwidth or quality of service. RTCP (Real Time Transport Control Protocol) is a control protocol used for quality of service. It is used for synchronisation of different media streams and to give information about participants in a

Multimedia applications have different requirements and there are many challenges to be addressed. On the whole, multimedia applications are divided into two types — interactive applications and non-interactive applications.

Interactive applications: Interactive applications are more time stringent. They are more sensitive to network jitter and end-to-end delay. Some loss of packets might be acceptable with such applications. Examples of such applications are Voice over IP, video conferencing apps, etc.

Non-interactive applications: These are less time stringent. They are less sensitive to jitter and delay, but a lesser loss of packets is to be expected. Applications like live streaming and stored video/audio fall into this category. Such applications have buffering on the client side to reduce network jitter and delay. Due to buffering, the video stream appears smooth. Many media players use such a design.

group session. RTP and RTCP use consecutive transport layer ports. RTP is mostly used with UDP/IP.

An RTP header would look like the one in Figure 1.

Raw data would look like what's shown in Figure 2.

V(version) 2bit: The first two bits are version identifiers.

P (padding) 1bit: If the padding bit is set, then the packets contain one or more padding bytes at the end. This is to make for a fixed header size.

X (extension) 1bit: This is mainly used by application developers if they need some separate information to be passed on to their clients.

M (marker) 1bit: Special frames are conveyed to the recipient through this bit. An example of this is the end frame which is marked.

PT (Payload Type) 7bits: PT identifies the type of payload. If there is any change in the data then the receptor knows it, and takes appropriate action. You can find the payload type in RTP RFC.

Sequence number 16bits: This is the number to put frames

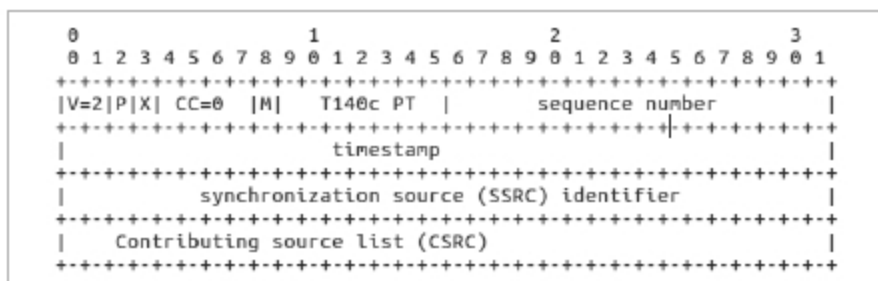


Figure 1: RTP header

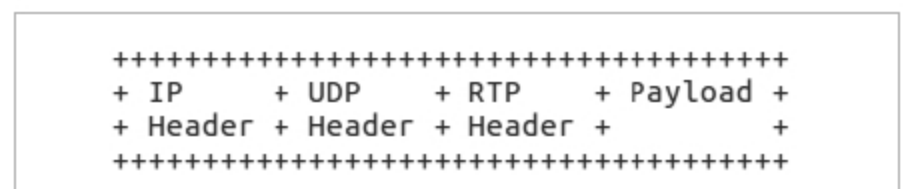


Figure 2: Raw data